INCH POUND
MIL-S-83519A
AMENDMENT 4
17 Feb 1993
SUPERSEDING
AMENDMENT 3
2 Nov 1990

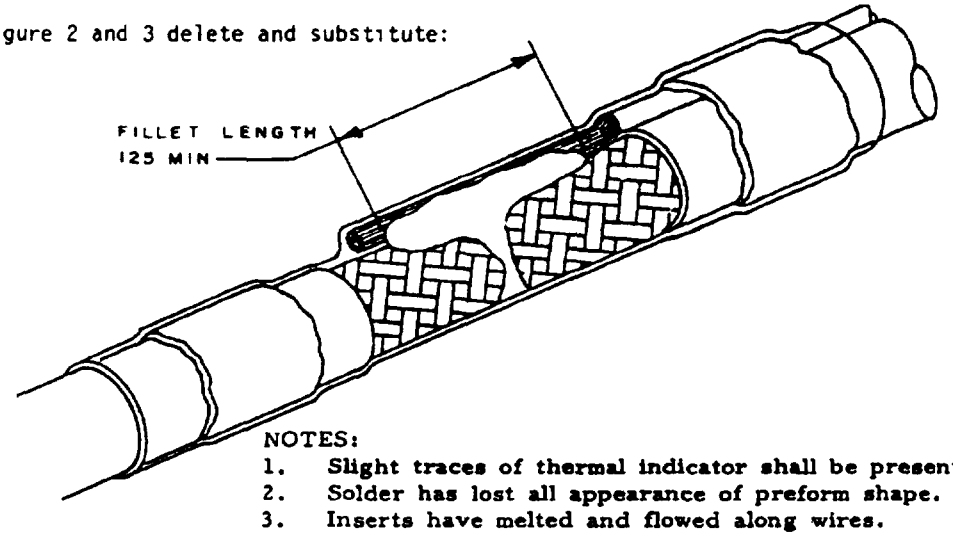
MILITARY SPECIFICATION

SHIELD TERMINATION, SOLDER STYLE,
INSULATED, HEAT-SHRINKABLE, ENVIRONMENT RESISTANT
GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification MIL-S-83519A, dated 21 June 1984, and is approved for use by all Departments and Agencies of the Department of Defense.

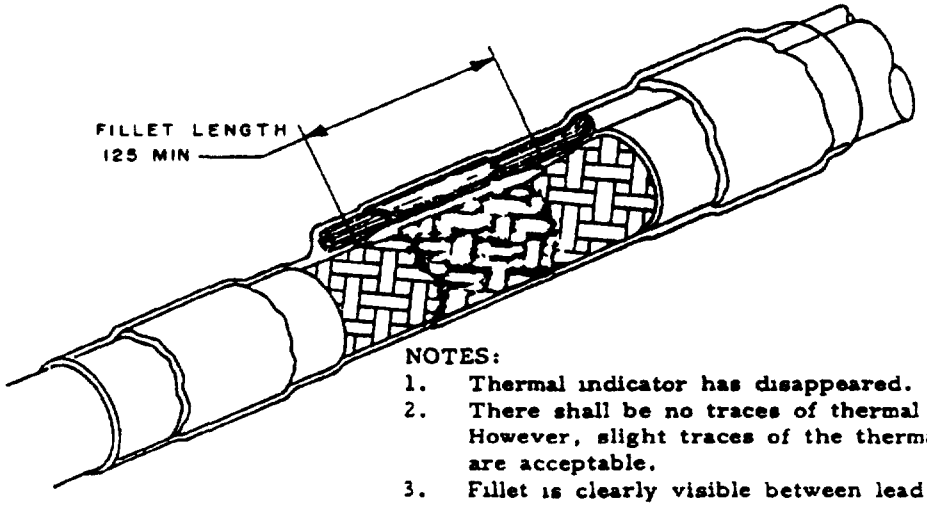
PAGE 6

Figure 2 and 3 delete and substitute:



NOTES:
1. Slight traces of thermal indicator shall be present.
2. Solder has lost all appearance of preform shape.
3. Inserts have melted and flowed along wires.
4. Shield and lead contours are visible.
5. There is a definite fillet visible along the lead and shield interface.

Figure 2. Acceptable termination. (maximum solder flow)



NOTES:
1. Thermal indicator has disappeared.
2. There shall be no traces of thermal indicator in the soldered area. However, slight traces of the thermal indicator in the insert area are acceptable.
3. Fillet is clearly visible between lead and shield.
4. Joint area is visible despite browning of sleeve.

Figure 3. Acceptable termination. (maximum solder flow)

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PAGE 2

* 2 1 Military.

delete MIL-C-25769 Cleaning Compound, Aircraft Surface, Alkaline Waterbase.

substitute MIL-C-87936 Cleaning Compounds, Aircraft Exterior Surfaces, Water Dilutable.

Standards.

delete DOD-STD-2000-3 Soldering Technology, High Quality/High Reliability.

substitute MIL-STD-2000 Standard Requirements For Soldered Electrical & Electronic Assemblies.

PAGE 3

3 3 3 delete and substitute·

3 3 3 Insulation sleeve The insulation sleeve shall be a transparent heat shrinkable, cross-linked thermoplastic The material must be suitable to meet all the performance requirements of this specification, when using a recommended heat source (see 6 2). The specific material shall be as specified on the applicable specification sheet

PAGE 7

3 7 delete and substitute

3 7 Marking Marking shall be in accordance with the requirements of MIL-STD-129 and other applicable DOD documents as well as appropriate federal statutes and regulations, e g , Federal Hazardous Substance Labeling Act, etc., unless otherwise prescribed by the contracting agency upon advice and recommendation of the departmental medical services

Following 3 7 add following new paragragh

3 8 Workmanship The integral components shall be held within the termination with sufficient force to withstand dislodging during normal installation. The termination shall be free of defects which could be detrimental to performance.

PAGE 9

TABLE I GROUP VIII delete and substitute

GROUP VIII (6 cable assemblies) 3 Type C, and 3 Type D assembly sets		
Visual and Dimensional	3.4 & 3.7	4.6.1
Fluid Immersion	3.5.2.5	4.7.5.4
Insulation Resistance	3.5.2.2	4.7.2
Dielectric Withstanding Voltage	3 5.2 3	4.7.3

PAGE 10

* 4.4.6 Retention of qualification.

delete· In addition to the activity responsible for qualification at any time during the 12 month period that the inspection data indicates failure of the qualified product to meet the requirements of the specification.

substitute· In addition to the periodic submission of inspection data, the QPL manufacturer shall immediately notify the qualifying activity at any time during the 12 month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification

* TABLE II.

Correct typographical error "ADL(% defective)" should read "AQL(% defective)".

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PAGE 11

* 4 5 3 2 Group C inspection.

delete in the order shown.

PAGE 12

4 6 3 delete and substitute·

4 6.3 Longitudinal change. Measure the length of the termination to the nearest .001 inch. Place terminations in clean aluminum dish and place in an oven at 200 ±2°C for 5 minutes. Remove from oven and allow to cool. When cool remeasure overall length of termination and calculate longitudinal change.

$$\% \text{ Change} = \frac{L_1 - L_2}{L_1} \times 100$$

L_1 = As supplied length
L_2 = Recovered length

PAGE 13

4 7.2 a. and b delete and substitute:

a Test condition· A
b Conditioning of assemblies Specimens shall be immersed as specified for 30 minutes prior to start of test (see 4 3 2)

4 7 3 a and b delete and substitute

a Conditioning of assemblies. Specimens shall be immersed as specified for 30 minutes prior to start of test (see 4.3.2).
b Magnitude and nature of potential. 1000 volts (RMS)

PAGE 14

TABLE V Test Condition c. delete and substitute:

c Place assembly in a 150°C ± 3°C oven for 22 hours

PAGE 15

Figure 4 delete and substitute

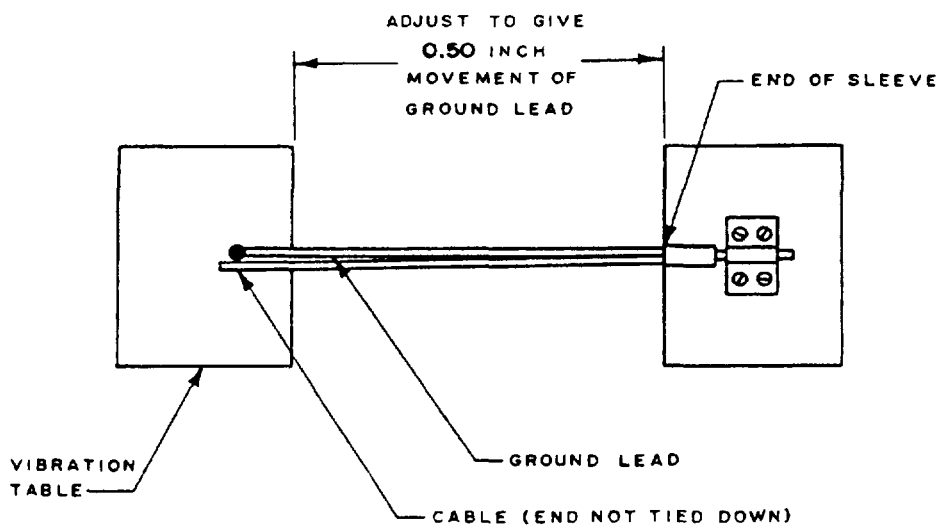


Figure 4 Shield termination

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PAGE 16

* 6 1 2 Thermal Indicator.

delete: DOD-STD-2000-3.

substitute MIL-STD-2000.

PAGE 17

6 6 delete and substitute

6 6 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Qualifying activity The activity responsible for the qualified products covered by this specification sheet is Naval Avionics Center, Code B/714, 6000 East 21st Street, Indianapolis, IN 46219-2189.

"The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment."

Custodians
Navy - AS
Air Force - 85
Army - ER

Review activities
Air Force - 99
Navy - SA, MC
Army - MI
DLA - GS

Preparing activity.
Air Force - 85

(Project No 5940-1145)

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